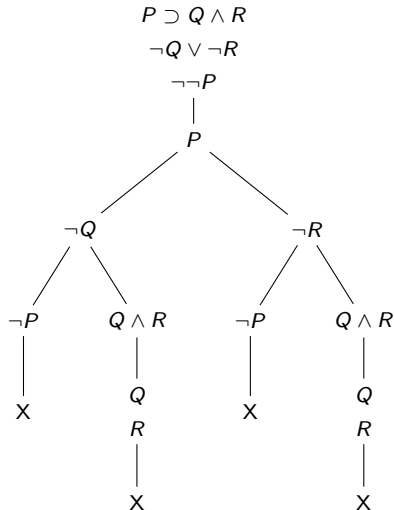


Solutions

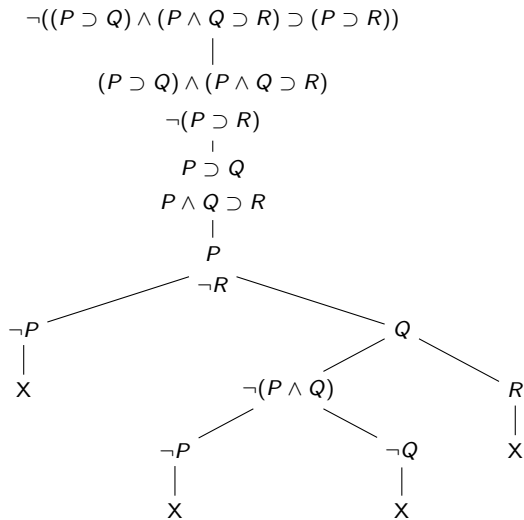


Note: different orderings of expansion rules are possible! But all lead to unsatisfiability.

Exercise

Check whether the formula $\neg((P \supset Q) \wedge (P \wedge Q \supset R) \supset (P \supset R))$ is satisfiable

Solution



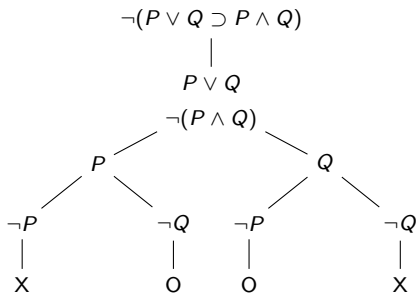
The tableau is closed and the formula is not satisfiable.

Satisfiability: An example

Exercise

Check whether the formula $\neg(P \vee Q \supset P \wedge Q)$ is satisfiable

Solution



Two open branches. The formula is satisfiable.

The tableau shows us all the possible interpretations $(\{P\}, \{Q\})$ that satisfy the formula.

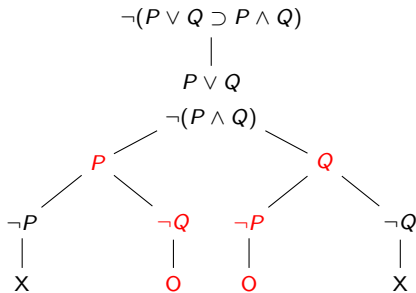
Using the tableau to build interpretations.

For each open branch in the tableau, and for each propositional atom p in the formula we define

$$\mathcal{I}(p) = \begin{cases} \text{True} & \text{if } p \text{ belongs to the branch,} \\ \text{False} & \text{if } \neg p \text{ belongs to the branch.} \end{cases}$$

If neither p nor $\neg p$ belong to the branch we can define $\mathcal{I}(p)$ in an arbitrary way.

Models for $\neg(P \vee Q \supset P \wedge Q)$



Two models:

- $\mathcal{I}(P) = \text{True}, \mathcal{I}(Q) = \text{False}$
- $\mathcal{I}(P) = \text{False}, \mathcal{I}(Q) = \text{True}$

Double-check with the truth tables!

P	Q	$P \vee Q$	$P \wedge Q$	$P \vee Q \supset P \wedge Q$	$\neg(P \vee Q \supset P \wedge Q)$
T	T	T	T	T	F
F	F	F	F	T	F
T	F	T	F	T	T
F	T	T	F	F	T

Homeworks!

Exercise

Show *unsatisfiability* of each of the following formulae using tableaux:

- $(p \equiv q) \equiv (\neg q \equiv p)$;
- $\neg((\neg q \supset \neg p) \supset ((\neg q \supset p) \supset q))$.

Show *satisfiability* of each of the following formulae using tableaux:

- $(p \equiv q) \supset (\neg q \equiv p)$;
- $\neg(p \vee q \supset ((\neg p \wedge q) \vee p \vee \neg q))$.

Show *validity* of each of the following formulae using tableaux:

- $(p \supset q) \supset ((p \supset \neg q) \supset \neg p)$;
- $(p \supset r) \supset (p \vee q \supset r \vee q)$.

For each of the following formulae, *describe all models* of this formula using tableaux:

- $(q \supset (p \wedge r)) \wedge \neg(p \vee r \supset q)$;
- $\neg((p \supset q) \wedge (p \wedge q \supset r) \supset (\neg p \supset r))$.

Establish the *equivalences* between the following pairs of formulae using tableaux:

- $(p \supset \neg p), \neg p$;
- $(p \supset q), (\neg q \supset \neg p)$;
- $(p \vee q) \wedge (p \vee \neg q), p$.

Termination

Assuming we analyse each formula at most once, we have:

Theorem (Termination)

For any propositional tableau, after a finite number of steps no more expansion rules will be applicable.

Hint for proof: This must be so, because each rule results in ever shorter formulas.

Note: Importantly, termination will *not* hold in the first-order case.

Soundness and Completeness

To actually believe that the tableau method is a valid decision procedure we have to prove:

Theorem (Soundness)

If $\Gamma \vdash \phi$ then $\Gamma \models \phi$

Theorem (Completeness)

If $\Gamma \models \phi$ then $\Gamma \vdash \phi$

Remember: We write $\Gamma \vdash \phi$ to say that there exists a closed tableau for $\Gamma \cup \{\neg\phi\}$.

Decidability

The proof of Soundness and Completeness confirms the decidability of propositional logic:

Theorem (Decidability)

The tableau method is a decision procedure for classical propositional logic.

Proof. To check validity of ϕ , develop a tableau for $\neg\phi$. Because of termination, we will eventually get a tableau that is either (1) closed or (2) that has a branch that cannot be closed.

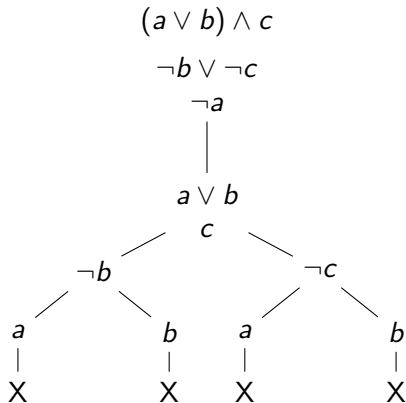
- In case (1), the formula ϕ must be valid (soundness).
- In case (2), the branch that cannot be closed shows that $\neg\phi$ is satisfiable (see completeness proof), i.e. ϕ cannot be valid.

This terminates the proof.

Exercise

Exercise

Build a tableau for $\{(a \vee b) \wedge c, \neg b \vee \neg c, \neg a\}$



Davis–Putnam–Logemann–Loveland (DPLL) procedure for satisfiability

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Revised and adapted by F. Patrizi

Decision procedures - automated reasoning

Automated reasoning is the discipline that aims at solving basic decision problems in logic, the main of which are:

Four types of decision problems

- **Model Checking**($\mathcal{I}, \phi$): $\mathcal{I} \stackrel{?}{\models} \phi$ Does the interpretation $\mathcal{I}$ satisfy ϕ , or equivalently, is $\mathcal{I}$ a model of ϕ ?
- **Satisfiability**(ϕ): $\stackrel{?}{\exists} \mathcal{I} . \mathcal{I} \models \phi$ Is there a model of ϕ ?
- **Validity**(ϕ): $\stackrel{?}{\models} \phi$ Is every interpretation for ϕ a model of ϕ ?
- **Logical implication**(Γ, ϕ): $\Gamma \stackrel{?}{\models} \phi$ Is every model of the set of formulas Γ a model of ϕ as well?
- **Logical inference**(Γ, ϕ): $\Gamma \stackrel{?}{\vdash}_{\Sigma} \phi$ Is there a proof of ϕ in Σ from Γ ?